INCH-POUND

MIL-C-3098/134C
24 September 1992
SUPERSEDING
MIL-C-3098/134B
8 March 1983

MILITARY SPECIFICATION SHEET

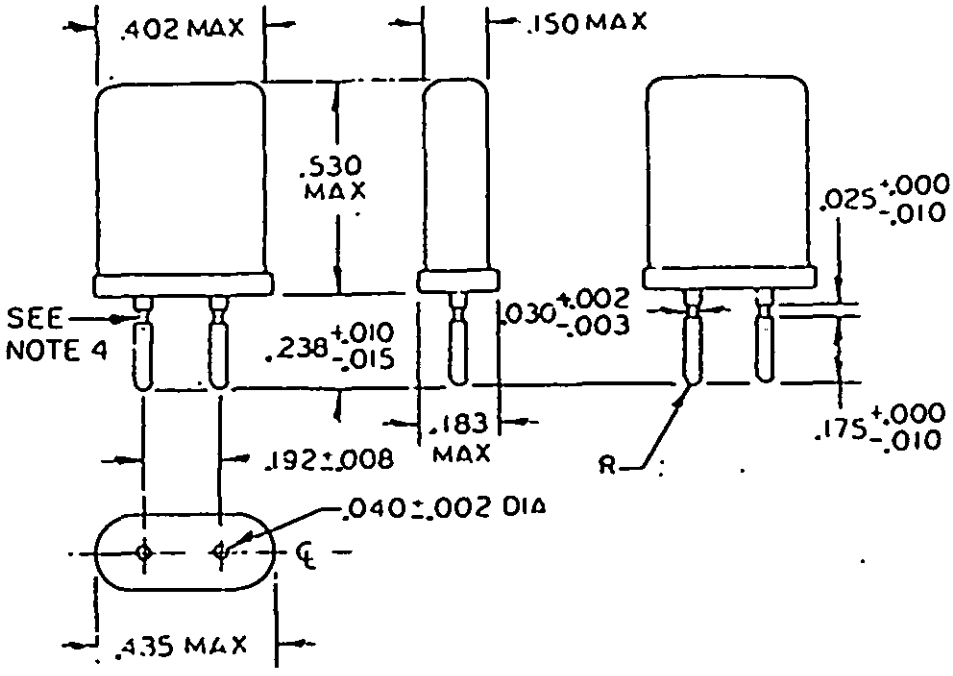
CRYSTAL UNIT, QUARTZ, CR153/U

This specification is approved for use by all Department and Agencies of the Department of Defense.

The requirement for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 10.0 to 30.0 MHz; fundamental; noncontrolled; antiresonance; nickel electroplated bonding.

Inches	mm
.002	0.05
.005	0.13
.008	0.20
.010	0.25
.015	0.38
.025	0.64
.030	0.76
.040	1.02
.150	3.81
.175	4.45
.183	4.65
.192	4.88
.238	6.05
.402	10.21
.435	11.05
.530	13.46



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.
4. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR153/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:59Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Holder: HC-42-2/U (MIL-H-10056/29). 1/

Dimensions, marking, and configuration: See figure 1.

Frequency range: 10.0 to 30.0 MHz, inclusive.

Frequency tolerance:

Operating temperature: ±10 ppm.

Equivalent resistance: 10 ohms maximum.

Mode of oscillation: Fundamental.

Temperature range:

Operable: 0°C to 60°C.

Operating: -55°C to +105°C, inclusive.

Drive level: 150 microwatts ±10 percent.

Antiresonance load capacitance: 20.0 ±0.1 pF.

C_1: 24 pF ±10 percent.

C_0/C_1 ratio: 245 ±20 percent.

Frequency stability $\frac{\Delta F}{F}$: ≤3 ppm over the temperature range 0°C to 60°C, when measured at series resonance.

Shock (specified pulse): 2/

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

1/ Mount shall be subminiature clip mount. Resonator shall be bonded into the subminiature clip mount by the nickel electroplated bonding method. Reference U.S. Army Command Technical Report ECOM 3452, dated August 1971, "Nickel Electrobonded Quartz Blanks".

2/ Method 213 of MIL-STD-202, test condition E, except duration shall be 11 milliseconds. Measurements before and after test. Frequency measured at +40°C ±1°C; and equivalent resonant resistance measured at 27°C ±1°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:59Z
Check the source to verify that this is the current version before use.

Aging:

Frequency change permitted: ±5 ppm.

QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SM
Navy - SH
Air Force - 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - ER

Agent
DLA - ES

(Project 5955-0672-07)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:59Z
Check the source to verify that this is the current version before use.